

Periodic Properties Dry Lab

1. On the blank periodic table, clearly indicate six (your choice) of the following element sections. You may use either color or letter codes.

a. Representative elements	b. Alkali metals
c. Transition elements	d. Alkaline earth metals
e. Inner-transition elements	f. Halogens
g. Chalcogens	h. Noble gases
i. Pnictogens	j. Noble metals
k. Metalloids	l. Lanthanide series
m. Post-transition metals	n. Actinide series

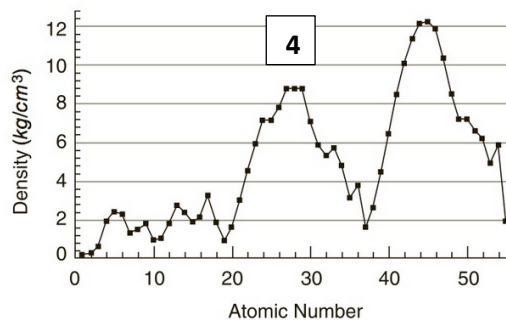
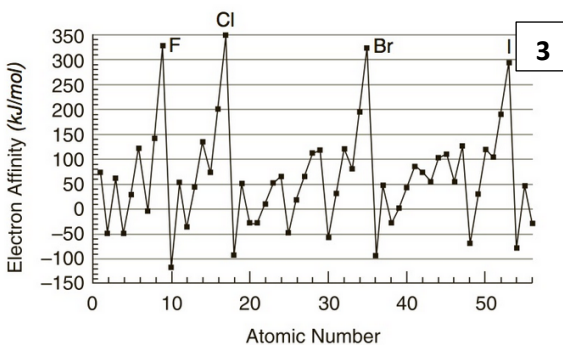
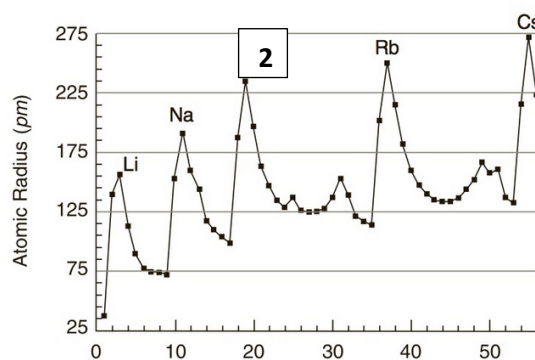
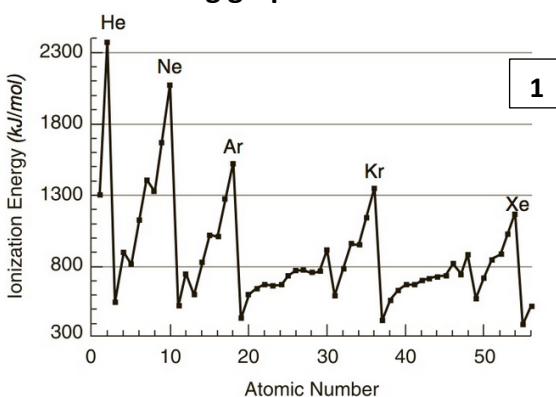
The image shows a 2D grid-based environment. The main area is a large grid with a central cyan-colored region. Within this cyan region is a smaller, darker brown rectangular area. To the left of the main grid, there are two vertical bars: a pink one and a green one. To the right, there is a blue vertical bar and a green vertical bar. Below the main grid, there are two horizontal rows of gray blocks. A small black dot is located in the upper left part of the main grid, and a small yellow dot is located in the bottom right row of gray blocks.

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2. Classify the following elements as using the classifications based on question 1. There may be more than one answer possible.

- Magnesium _____
- Plutonium _____
- Argon _____
- Zr _____
- Bromine _____
- K _____
- Element 118 _____
- Pb _____
- Titanium _____

Use the following graphs to answer the rest of the questions.



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3. Refer to Figure 1. Circle the option of the pair with the highest ionization energy.
 - a. C or O
 - b. Mg or Al
 - c. P or S
 - d. Mg or Ca
4. Refer to Figure 2. Circle the option of the pair with the smallest atomic radius.
 - a. C or O
 - b. Mg or Al
 - c. P or S
 - d. Mg or Ca
5. Based on your answers to 3 and 4, what generalization can be made regarding how ionization energy is related to atomic size?

As an atom's radius gets smaller, its ionization energy goes up.

6. Based on Figure 3, what is the general periodic trend for electron affinities?

It could possibly be a linear trend. It is hard to tell (to me) really.

7. Based on Figure 3, which group has the highest electron affinity?

Group 17.

8. Based on Figure 3, which element has the highest electron affinity?

It appears that chlorine has the highest electron affinity, based on Fig.3.

9. Based on Figure 3, how does electron affinity relate to atomic size (Figure 2)?

The smaller the size of the atom, the more electron affinity it has.

10. Based on Figure 4, what is the trend in density as you go down a family?

It looks like an exponential trend.

11. Based on Figure 4, what is the trend in density for periods 2 and 3?

During period 2, it seems to go from 2 up to almost 3, then decreases back down towards one. Negative-Linear. Period 3 is parabolic.

12. Based on Figure 4, which element has the greatest density?

Element 46, Palladium.